Introduction to SINDy

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1 Introduction

Traditionally, engineers use established theories to write the equations for physical systems. Today, modern data-driven tools allow us to discover these governing equations directly from raw measurements. This report explores two powerful algorithms designed for this exact task: SINDy (Sparse Identification of Nonlinear Dynamics) and PySR (Python Symbolic Regression).

SINDy works by creating a large library of possible mathematical terms, such as standard variables, derivatives, and polynomials. It then uses a filtering technique to throw away the unnecessary terms, leaving behind only the simplest equation that explains the data. PySR takes a different approach. Instead of a pre-made library, it uses an evolutionary algorithm to randomly combine, mutate, and test mathematical symbols until it builds an expression that fits the system's behavior.

While both tools are incredibly powerful, using them on real-world data presents a major challenge. If an algorithm is given too much mathematical freedom, it will often try to fit the random noise in the measurements rather than the actual physics. This results in equations that are overly complicated and mathematically unstable over time. In this paper, we apply these methods to classic physical problems—specifically electrical circuits and spatial heat distribution—to investigate these challenges and explore what it takes to extract true, reliable physical laws from raw data.

2 Background: Sparse and Symbolic Regression

SINDy. Sparse Identification of Nonlinear Dynamics (SINDy) [1] aims to recover governing equations by selecting a sparse set of active terms from a predefined library of candidate functions. Given data $\mathbf{x}(t)$, a library $\Theta(\mathbf{x})$ is constructed (e.g., polynomials, trigonometric functions, interactions), and sparse regression—typically LASSO or sequential thresholded least squares [**?**]—is used to solve:

$$\dot{\mathbf{x}} = \Theta(\mathbf{x})\,\boldsymbol{\xi}, \tag{1}$$

where $\boldsymbol{\xi}$ is constrained to be sparse [1]. The result is a minimal, interpretable model that explains the observed dynamics without requiring prior knowledge of the governing equations. *Example:* For a nonlinear oscillator, SINDy may recover $\dot{x} = ax + bx^3$, selecting only the dominant terms from a larger candidate set.

Symbolic Regression (PySR). Symbolic regression [3] seeks an explicit functional form $\hat{y} = f(\mathbf{x})$ by searching over the space of mathematical expressions. PySR performs this search using evolutionary optimization [**?**], iteratively generating and refining candidate equations while balancing accuracy and complexity via a Pareto front:

$$\min_{f} \; \mathcal{L}\big(y,\, f(\mathbf{x})\big) + \lambda \cdot \text{Complexity}(f), \tag{2}$$

where simpler expressions are favored [3]. Unlike SINDy, no predefined library is required; the search explores combinations of elementary operators and functions. *Example:* Given input–output data, PySR may recover $\hat{y} = x^2 + \sin(x)$ from a broad, unconstrained search.

Summary. SINDy performs sparse selection within a fixed, user-defined library and is especially suited to dynamical systems where the derivative structure is known [1]. Symbolic regression explores a flexible, open-ended expression space and is better suited to discovering static input–output relationships [3]. Both methods prioritize interpretability and compact representations, making them attractive alternatives to black-box models in scientific and engineering applications [**?**].

3 RLC circuit

In this section, we will use SINDy in order to retrieve an RLC circuit's parameters: R, L, and C. In addition, we will find the differential equation that represents the resistor's voltage at all times in the circuit, from the data V(t). At t=0, the switch opens and we get a parallel RLC circuit.

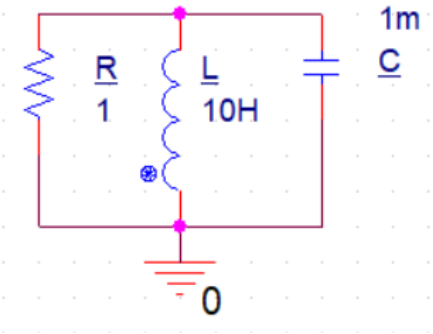


Figure 1: Parallel RLC circuit

3.1 The circuit

The chosen values for R, L, and C are as shown in Figure 1. In addition, we chose an initial condition of $1V$ and assumed that the current through the inductor is 0 as an initial condition. In other words:

$$V_R(t=0) = 1, \quad I_L(t=0) = 0$$

The differential equation we expect to get from SINDy is:

$$0 = \frac{d^2V}{dt^2} + 1000\frac{dV}{dt} + 100V$$

3.2 The graph

This is the graph of $V_R(t)$ from $t = 0$ until the circuit stabilizes, arbitrarily choosing an end time of $t = 10ms$. This graph will be exported as a CSV file and will be the input of our code:

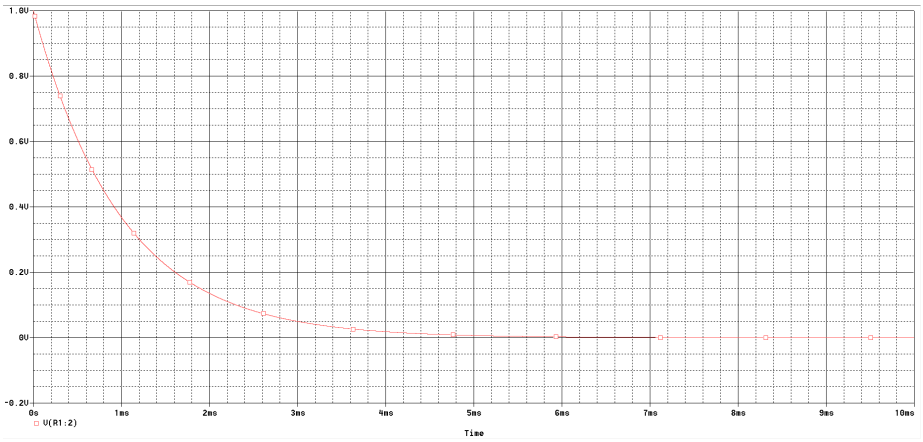


Figure 2: Voltage drop on R

3.3 Reconstructing the differential equation - the simplified method

In the simplified method of reconstructing the differential equation from our data, we let SINDy know exactly what our differential equation should look like $\frac{d^2V}{dt^2} = a\frac{dV}{dt} + bV$, and its only mission was to find the coefficients a, b. In our case, the expected theoretical values were $a = -1000$ and $b = -100$. SINDy proved successful in this challenge, yielding $a = -1000$ and $b = -100.214$. In conclusion, when many data points are available and the exact structure of the differential equation is known, SINDy has proven to be a reliable tool for accurately reconstructing differential equations.

3.4 Reconstructing the differential equation - Unconstrained Discovery method

In the unconstrained discovery method, we did not provide SINDy with information about the structure of the equation we needed to reconstruct, allowing it to consider any order of derivatives, polynomials, and combinations thereof. Since we are assuming we do not know the differential equation we are aiming for, we will assume arbitrarily that it is of order 3. This is the more relevant method for real-world problems, for which we do not know what structure of differential equation to expect, such as wet antennas. SINDy yielded the following:

$$\begin{aligned}\frac{d^3V}{dt^3} = &-2347991.510 + 255137.701V + 178329.611\frac{d^2V}{dt^2} + 6921580.204t \\ &- 3216.684V^2 + 10125.742V\frac{d^2V}{dt^2} + 73451.616Vt - 11328.552\left(\frac{d^2V}{dt^2}\right)^2 \\ &- 2041126.306\frac{d^2V}{dt^2}t - 6983528.813t^2 - 3308.155V^3 - 108262.030V^2\frac{d^2V}{dt^2} \\ &- 2809.992V^2t + 201321.098V\left(\frac{d^2V}{dt^2}\right)^2 - 119994.481V\frac{d^2V}{dt^2}t + 64429.232Vt^2 \\ &- 115418.338\left(\frac{d^2V}{dt^2}\right)^3 + 97166.441\left(\frac{d^2V}{dt^2}\right)^2 t + 4146542.573\frac{d^2V}{dt^2}t^2 + 2399216.917t^3\end{aligned}$$

Obviously, this result is extremely different from the theoretical equation we expected SINDy to produce. The differential equation is of order 3; there are many more terms than expected, and the structure itself is completely unlike our expected equation. Note that there are no $\frac{dV}{dt}$ terms, since SINDy had concluded that $\frac{dV}{dt} = 0$. Our mission is now to filter out non-physical terms and develop a method to decide whether each term is necessary.

3.4.1 Minimizing the options given to SINDy

In order to ensure the non-physical terms are not included in the differential equation, we will not give SINDy the option to use them from the start. In our case, we do not want anything raised to any power higher than 1, and we do not want a constant. These changes alone yield the new differential equation:

$$\frac{d^3V}{dt^3} = -4990.182V + -499.000\frac{dV}{dt} + -10.002\frac{d^2V}{dt^2}$$

Although not perfect, we made a lot of progress in getting closer to our desired equation. Note that SINDy no longer concluded that $V' = 0$. We got a physical equation with reasonable coefficients and decent terms, though it is of order 3 instead of order 2. Plotting the solution to the yielded equation, as shown in Figures 3 and 4, the differences are very clear. The theoretical equation decays very quickly, while the yielded equation does not decay at all, but oscillates around 0 instead.

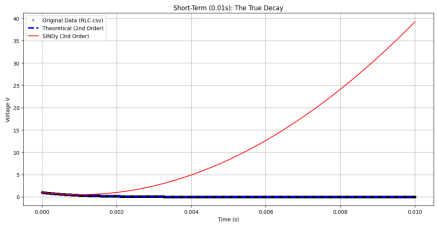


Figure 3: Theoretical and yielded result - short term comparison

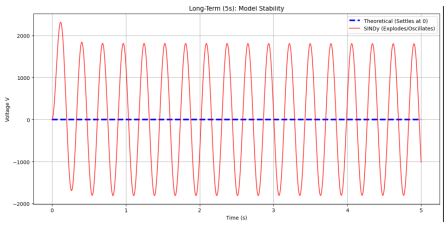


Figure 4: Theoretical and yielded result - long term comparison

3.4.2 Perfecting the unconstrained discovery method

Despite the progress made in filtering terms and powers in the previous subsection, we were still quite far from the required equation. The oscillations in figure 4 was a clear sign of overfitting, which had lead us to understand that we had to lower the order of the differential equation from 3 to 2. Using the exact same code with the small change of lowering the differential equation order, we get the result:

$$0 = \frac{d^2V}{dt^2} + 1000\frac{dV}{dt} + 100.203V$$

This result is extremely close to the theoretical equation of the RLC circuit. In the end, we essentially returned to the simpler method and achieved a result that perfectly matches our expectations. We started this process by treating the system as a "black box," meaning we pretended to know absolutely nothing about its underlying physics or the correct order of the differential equation. Because we gave the algorithm so much freedom, our initial models were too complex and tried to fit the random noise in the data instead of the real physics. However, through a careful process of filtering terms, adjusting the model's settings, and testing the equations for long-term stability, we were able to remove the incorrect parts. This proves that even when we start with zero background knowledge, we can still guide the SINDy algorithm to discover the true physical laws of a system directly from the raw data.

4 Heat-equation PDE discovery

Abstract

We study sparse data-driven discovery of the one-dimensional heat equation under controlled measurement noise. The objective is not only to recover the diffusion coefficient, but also to determine whether the identified PDE structure remains physically meaningful as noise increases. To this end, we compare a direct pointwise baseline based on `PDELibrary` with a weak-form alternative based on `WeakPDELibrary`. Model quality is assessed through four complementary diagnostics: the estimated diffusion coefficient as a function of noise, mean coefficient heatmaps,

term-selection frequency heatmaps, and a term-wise robustness metric denoted TRI (Term Robustness Index). This framework turns the qualitative question of which discovered terms should be trusted into an explicit and reproducible analysis procedure.

4.1 Problem formulation

We consider the one-dimensional heat equation

$$u_t = \kappa u_{xx}, \qquad x \in [0,1], \quad t \in [0,1], \tag{3}$$

with homogeneous Dirichlet boundary conditions

$$u(0,t) = u(1,t) = 0, \tag{4}$$

and initial condition

$$u(x,0) = a_1 \sin(\pi x) + a_2 \sin(2\pi x), \tag{5}$$

where $\kappa = 0.1$, $a_1 = 1$, and $a_2 = 0.5$. For this choice, the exact solution is

$$u(x,t) = a_1 \sin(\pi x) e^{-\kappa \pi^2 t} + a_2 \sin(2\pi x) e^{-4\kappa \pi^2 t}. \tag{6}$$

This benchmark is intentionally simple. The true PDE contains a single physically meaningful spatial derivative term, namely u_{xx}. Consequently, the repeated appearance of additional terms can be interpreted directly as a loss of robustness or as an artifact induced by noise and model flexibility.

The synthetic data are sampled on a uniform grid with $n_x = 101$ spatial points and $n_t = 200$ temporal points. Additive Gaussian noise is injected into the state measurements,

$$u^\eta(x,t) = u(x,t) + \varepsilon(x,t), \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2), \tag{7}$$

using the variance set

$$\sigma^2 \in \left\{0,\, 5\times10^{-5},\, 10^{-4},\, 2\times10^{-4},\, 5\times10^{-4},\, 10^{-3},\, 2\times10^{-3},\, 5\times10^{-3}\right\}. \tag{8}$$

Each noise level is repeated three times. The same noise realizations are used for both identification methods in order to preserve a fair one-to-one comparison.

Table 1: Key settings for the heat-equation discovery experiments.

True PDE	$u_t = 0.1\, u_{xx}$
Spatial-temporal domain	$x \in [0,1]$, $t \in [0,1]$
Grid resolution	$n_x = 101$, $n_t = 200$
Initial condition	$\sin(\pi x) + 0.5 \sin(2\pi x)$
Noise variances	0 to 5×10^{-3} (8 levels)
Runs per noise level	3
Sparse optimizer	STLSQ
Maximum derivative order	2

4.2 Identification strategy

Two SINDy-based PDE discovery pipelines are compared.

Raw pointwise baseline. The first model uses `PDELibrary`, polynomial terms up to degree 2, and spatial derivatives up to order 2. This is the direct pointwise approach. In the present implementation, the raw baseline is supplied with the exact time derivative u_t in order to isolate the effect of library construction from errors caused by numerical differentiation in time. This makes the baseline intentionally favorable. If it still degrades under noise, the failure can be attributed primarily to the pointwise formulation rather than to time-derivative estimation alone.

Weak-form noise-robust model. The second model uses `WeakPDELibrary` with the same polynomial degree and derivative order, but identifies the PDE in weak form by integrating against localized test functions. The weak formulation is designed to reduce direct amplification of measurement noise during differentiation and is therefore expected to retain the physically correct diffusion term more reliably as the noise level increases.

Both methods use sparse thresholded least squares (STLSQ) to select a parsimonious representation from the candidate library. The main comparison is therefore not between two unrelated learning algorithms, but between two representations of the same sparse-identification principle: direct pointwise regression and weak-form regression.

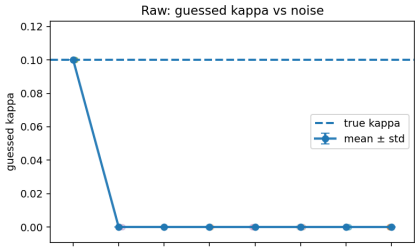
4.3 Robustness diagnostics

A single recovered equation is not sufficient to evaluate discovery quality under noise. We therefore use four complementary diagnostics.

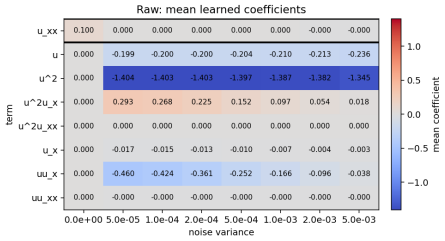
Estimated diffusion coefficient. For each run, the coefficient multiplying u_{xx} is interpreted as the identified diffusion coefficient $\hat{\kappa}$. A robust method should keep $\hat{\kappa}$ close to the true value $\kappa = 0.1$ while exhibiting limited variation across repeated realizations.

Mean coefficient heatmaps. For each candidate term and each noise level, we compute the mean learned coefficient over repeated runs. Displaying these values as a heatmap makes it possible to assess whether the correct term u_{xx} remains dominant and whether spurious terms emerge systematically or only intermittently.

Figures 5 and 6 present these two diagnostics together for each identification strategy. In each case, the left panel shows the recovered diffusion coefficient as a function of noise variance, and the right panel shows the corresponding mean coefficient heatmap over the full candidate library.

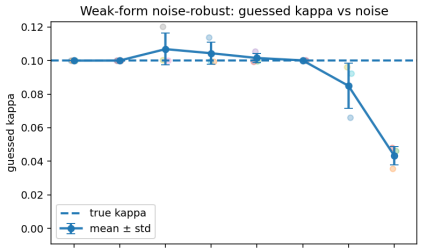


(a) Identified $\hat{\kappa}$ as a function of noise variance.

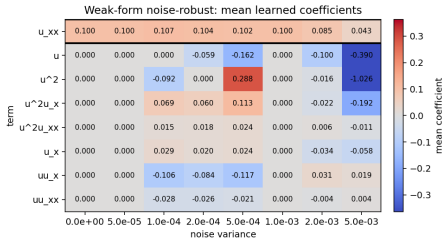


(b) Mean learned coefficients across noise levels.

Figure 5: Raw pointwise baseline. Left: recovered diffusion coefficient as a function of noise variance. Right: mean coefficient heatmap across candidate terms and noise levels.



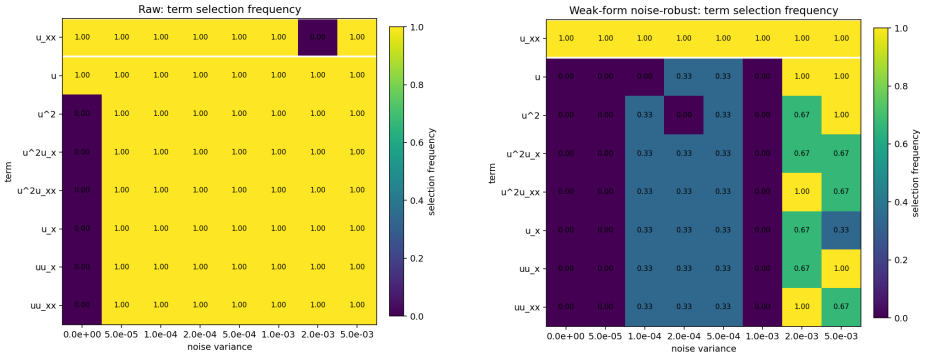
(a) Identified $\hat{\kappa}$ as a function of noise variance.



(b) Mean learned coefficients across noise levels.

Figure 6: Weak-form noise-robust model. Left: recovered diffusion coefficient as a function of noise variance. Right: mean coefficient heatmap across candidate terms and noise levels.

Selection-frequency heatmaps. Mean coefficients alone do not distinguish between a term that is consistently selected with moderate amplitude and a term that appears sporadically with large magnitude. For that reason, we also record the selection frequency of each term, defined as the fraction of runs in which the corresponding coefficient exceeds a small magnitude threshold. These heatmaps reveal whether a term is persistently present or merely noise-driven.



(a) Raw pointwise baseline: term-selection frequency across noise levels.

(b) Weak-form model: term-selection frequency across noise levels.

Figure 7: Selection-frequency heatmaps for the two identification methods.

4.4 Term Robustness Index (TRI)

The previous diagnostics provide a detailed visual picture, but PDE discovery under noise still leaves a practical question unresolved: when several candidate terms appear, which of them should be regarded as physically credible. A useful ranking should reward terms that appear repeatedly, preserve the same sign, retain a stable magnitude as noise varies, and remain present in the weak-form model that is explicitly designed to be noise-robust. To formalize this idea, we define the *Term Robustness Index* (TRI).

Consider a candidate term q. Let R_q and W_q denote its average selection frequencies in the raw and weak-form methods, respectively. Here, selection frequency means the fraction of runs in which the coefficient of the term exceeds a small magnitude threshold. TRI is built from four components.

1. **Presence consistency** (P)**.** A robust term should appear repeatedly across runs, across noise levels, and across both discovery pipelines. We define
$$P_q = \frac{1}{2}\left(R_q + W_q\right). \tag{9}$$
This quantity is large when the term is selected often in both methods.

2. **Sign consistency** (S)**.** A physically meaningful term should not switch sign unpredictably from one run to another. Let n_+ and n_- denote the numbers of nonzero positive and nonzero negative instances of the term over all runs and both methods. We define
$$S_q = \begin{cases} \dfrac{\max\{n_+, n_-\}}{n_+ + n_-}, & n_+ + n_- > 0, \\ 0, & n_+ + n_- = 0. \end{cases} \tag{10}$$
Thus, S_q is close to 1 when the term keeps the same sign whenever it is active.

3. **Magnitude stability** (M)**.** Presence and sign consistency do not by themselves guarantee that a term is reliable. A term may be selected often and keep the same sign, yet still receive coefficient values whose magnitude changes substantially as the noise level increases. We therefore introduce a separate measure that quantifies how stable the coefficient magnitude remains across the noise study.

   For one identification method, let

$$m_1, \ldots, m_r$$

   denote the mean coefficient values of the term across the r tested noise levels, after discarding values whose magnitude falls below the selection threshold. Since sign consistency is already measured separately by S_q, we assess only the stability of the coefficient magnitude and therefore consider

$$|m_1|, \ldots, |m_r|.$$

   For a single method, we define the stability measure

$$\text{Stability} = \frac{1}{1 + \text{CV}}, \qquad \text{CV} = \frac{\text{std}(|m_1|, \ldots, |m_r|)}{\text{mean}(|m_1|, \ldots, |m_r|) + \varepsilon}, \tag{11}$$

   where ε is a small positive constant introduced only to avoid division by zero. This quantity is close to 1 when the coefficient magnitude remains nearly constant across noise levels, and decreases toward 0 as the magnitude fluctuates more strongly.

   Applying this definition to the raw and weak pipelines yields two method-specific stability values,

$$M_{q,\text{raw}} = \text{Stability(raw mean coefficients)}, \qquad M_{q,\text{weak}} = \text{Stability(weak mean coefficients)}. \tag{12}$$

   The TRI magnitude component is then defined as the average of these two method-specific quantities,

$$M_q = \frac{1}{2}\left(M_{q,\text{raw}} + M_{q,\text{weak}}\right). \tag{13}$$

   In other words, $M_{q,\text{raw}}$ and $M_{q,\text{weak}}$ are the raw-method and weak-method versions of the same stability measure, while M_q is their combined contribution to TRI.

4. **Denoising or method survival** (D)**.** The final component explicitly rewards persistence in the weak-form model, which is the noise-robust method under study. Using the same notation as above, we define

$$D_q = W_q \left(0.5 + 0.5\frac{W_q}{R_q + W_q + \varepsilon}\right), \tag{14}$$

   where ε is again a small positive constant. This term is large when the candidate survives strongly in the weak-form method, especially when that survival is not matched by the raw baseline.

The complete Term Robustness Index is the weighted combination

$$\mathrm{TRI}_q = 0.25P_q + 0.20S_q + 0.20M_q + 0.35D_q. \tag{15}$$

The larger weight assigned to D_q reflects the central purpose of the experiment, namely to identify terms that remain reliable specifically under the weak-form, noise-robust formulation.

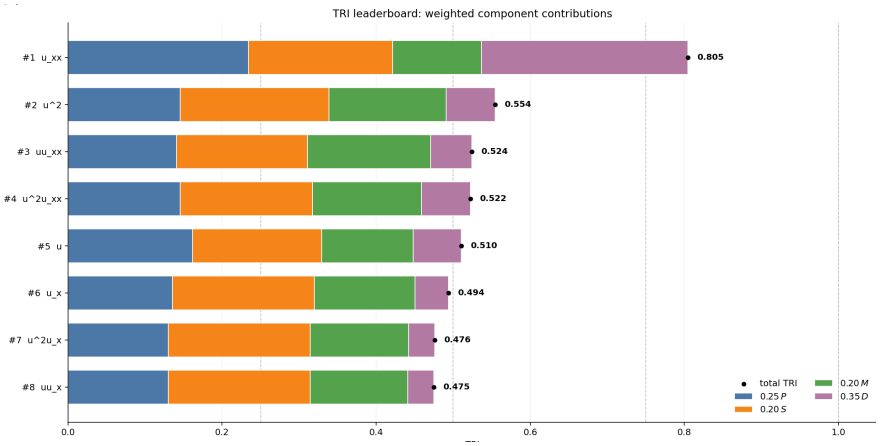


Figure 8: TRI leaderboard. Terms are ranked by the Term Robustness Index, with bar segments showing the weighted contributions of the individual components P, S, M, and D.

4.5 Interpretation of the diagnostics

The four diagnostics are intended to be read together rather than in isolation. A successful recovery of the heat equation should satisfy the following qualitative pattern:

1. The identified diffusion coefficient $\hat{\kappa}$ remains close to $\kappa = 0.1$ over the full noise sweep.
2. The term u_{xx} remains visually dominant in the mean coefficient heatmaps.
3. Spurious terms have low or inconsistent selection frequency.
4. u_{xx} appears at or near the top of the TRI leaderboard.

This multi-view analysis is more informative than reporting a single discovered equation at a single noise level. In noisy settings, a term may appear plausible in one realization and disappear in the next. Robust discovery therefore requires persistence across repeated trials, stable sign, limited coefficient drift, and survival under the weak formulation. TRI summarizes these requirements in a compact ranking, while the heatmaps preserve the underlying structure required for interpretation.

4.6 Conclusion

The present workflow turns PDE discovery for noisy heat-equation data into a controlled robustness study rather than a one-shot regression exercise. The raw pointwise library is retained as a baseline because it exposes the failure modes of direct differentiation, while the weak-form library provides the noise-robust alternative of primary interest. The combination of $\hat{\kappa}$-versus-noise plots, coefficient heatmaps, optional selection-frequency heatmaps, and the Term Robustness Index yields a concise but informative framework for deciding which discovered terms are physically credible. In this setting, the objective is not merely to fit the data, but to identify a PDE structure that remains interpretable and stable as measurement quality degrades.

5 Opportunistic Weather-Retrieval with Wireless Communication Networks

5.1 CMLs as Opportunistic Sensors

Commercial Microwave Links (CMLs) in cellular networks are highly sensitive to atmospheric conditions, inadvertently acting as dense, near-ground weather sensors [7]. Raindrops scatter and absorb microwave signals, causing measurable signal attenuation. The fundamental principle of using wireless links for rainfall retrieval relies on path-integrated attenuation, as conceptually diagrammed in 9.

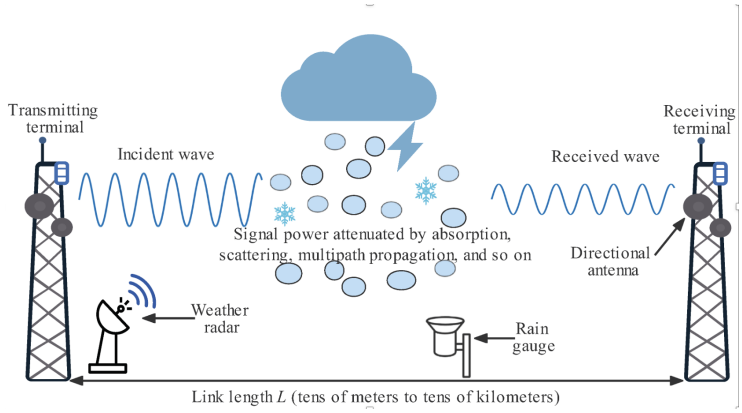


Figure 9: Conceptual diagram of mmWave signal propagation affected by path-integrated rainfall attenuation. The diagram demonstrates path-integrated attenuation caused by rainfall distributed along the link path.

Extracting Attenuation. Networks continuously monitor Transmitted and Received Signal Levels (TSL and RSL). The total rain-induced attenuation,

$A(t)$ (in dB), is:

$$A(t) = TSL(t) - RSL(t) - A_{baseline} \tag{16}$$

where $A_{baseline}$ is the clear-sky attenuation (which includes free-space path loss and gaseous absorption). This baseline is typically established by averaging RSL during known dry periods [9]. The magnitude of attenuation scales with physical link length and is significantly more severe at higher frequencies (e.g., 38 GHz vs. 18 GHz) [4].

5.2 Physical Background: Rain-Induced Attenuation

CML-based rain estimation relies on the ITU Power Law. Because the instantaneous rain rate $R(l,t)$ varies along the propagation path l, the total attenuation over a link of length L (in km) is approximated using a path-averaged rain rate, $\bar{R}(t)$:

$$A(t) = \int_0^L aR(l,t)^b dl \approx a\bar{R}(t)^b L \tag{17}$$

where a and b are empirical constants dependent on frequency, polarization, and temperature.

Power-Law Estimation.
By inverting this equation, the baseline estimate for the path-averaged rain rate, $\hat{R}_{pl}(t)$, can be derived directly from the measured attenuation:

$$\hat{R}_{pl}(t) = \left(\frac{A_r(t)}{aL}\right)^{\frac{1}{b}} \tag{18}$$

Wet Antenna Attenuation (WAA). A key source of systematic error in CML-based rainfall estimation is wet antenna attenuation (WAA), the additional signal loss caused by water films forming on antenna radomes during and after rainfall events [10]. Schleiss et al. [10] quantified WAA on a 1.85 km link at 38 GHz and proposed a dynamic exponential model (SRB model) with the following properties: WAA is bounded by a maximum of $A_{\max} \approx 2.3$ dB, rises exponentially during rainfall, and decays exponentially after rain ceases.

Formally, let $A_w(t)$ denote the WAA (in dB) at time t. During a rain event starting at t_0:

$$A_w(t) = A_{\max}\Big(1 - e^{-(t-t_0)/\tau_{\text{wet}}}\Big), \tag{19}$$

and during the subsequent drying phase starting at t_1:

$$A_w(t) = A_w(t_1)\, e^{-(t-t_1)/\tau_{\text{dry}}}, \tag{20}$$

where τ_{wet} and τ_{dry} are the wetting and drying time constants, respectively, both calibrated from link measurements alone [10].

5.3 Weather-Induced Attenuation Model

The total observed attenuation $A_{\text{obs}}(t)$ aggregates multiple physical contributions [2]:

$$A_{\text{obs}}(t) = A_r(t) + A_{\text{wa}}(t) + A_{\text{base}}(t) + A_{\text{other}}(t) + n(t). \tag{21}$$

Each component is commonly interpreted as follows:

1. $A_r(t)$ denotes the rain-induced path attenuation. It is commonly approximated using the ITU power-law, i.e., $A_r(t) \approx aLR(t)^b$, where $R(t)$ is the path-averaged rain rate and a, b are frequency- and polarization-dependent coefficients [8, 4]. This formulation is static and instantaneous, and assumes spatially uniform rainfall along the link. Its accuracy may degrade for long links, low rain rates, or spatially heterogeneous precipitation.

2. $A_{\text{wa}}(t)$ denotes wet-antenna attenuation, namely the excess loss caused by water accumulation on the antenna radome. Existing models often describe this term using exponential rise and decay dynamics [10], reflecting its memory effect, since attenuation may persist after rainfall ceases. The corresponding parameters, such as τ_{wet}, τ_{dry}, and $A_{\max}$, are link- and climate-specific and are generally not directly observable.

3. $A_{\text{base}}(t)$ denotes the slowly varying baseline attenuation associated with dry-weather propagation and hardware drift. In practice, this component is typically estimated and removed during preprocessing [2], under the assumption that it varies slowly relative to rainfall dynamics. This assumption may become less accurate during long events or rapid environmental changes.

4. $A_{\text{other}}(t)$ denotes additional environmental attenuation contributions, such as fog, humidity, or dust [5]. During rain events at operational frequencies, these effects are typically assumed to be small relative to rain-induced attenuation and are therefore neglected or absorbed into the residual term.

5. $n(t)$ denotes random measurement noise, including quantization effects and hardware fluctuations [2]. This component is commonly reduced through smoothing or filtering prior to numerical differentiation, although residual noise may still affect the dynamic estimates.

Simplified Model. Under these assumptions, the working model reduces to

$$\tilde{A}(t) \approx A_r(t) + A_{\text{wa}}(t), \tag{22}$$

where $\tilde{A}(t) = A_{\text{obs}}(t) - A_{\text{base}}(t)$ denotes the baseline-corrected attenuation, which serves as the input to the SINDy identification framework.

5.4 Goals: Data-Driven Discovery of Attenuation Dynamics

5.4.1 Goal 1: Rain-Induced Attenuation Relationship.

We seek to discover the static relationship $\tilde{A}(t) \approx f(R(t))$ from CML data using PySR, without imposing the ITU power-law a priori. The ITU model aLR^b serves as both a validation benchmark and a structural reference for constrained PySR experiments.

1a. ($A_w \approx 0$): PySR is applied over rain periods neglecting wet-antenna effects, discovering the purest possible $f(R)$ from $\tilde{A}(t)$ directly.

1b. ($A_w \not\approx 0$): WAA is accounted for by subtracting an SRB estimate, such that $\tilde{A}(t) - \hat{A}_w^{\mathrm{SRB}}(t) \approx A_r(t) \approx f(R(t))$. Comparing 1a and 1b quantifies the bias introduced by unmodeled wet-antenna attenuation on the discovered relationship.

Research Questions.

RQ1 Is the power-law aLR^b the correct functional form of the rain-attenuation relationship, or does PySR recover a different $f(R)$?

- **RQ1a** Does the discovered $f(R)$ change between stage 1a ($A_w \approx 0$) and stage 1b (A_w corrected)?

RQ2 Does unmodeled WAA corrupt the discovered $f(R)$, introducing additional terms beyond the power-law?

- **RQ2a** Does SRB subtraction simplify the discovered $f(R)$ toward aLR^b — suggesting the extra terms in 1a are WAA artifacts rather than rain physics?

5.4.2 Implementation: Rain-Induced Attenuation.

Goal: Create Dataset (Using Meta and Temporal Raw Data). Meaning: Prepare Links and RGs, etc.
Input: Raw CML data, metadata, and unaligned reference data.
Output: Prepared dataset for regression.
Data Preparation: CML-RG Alignment.

1. **Spatial and Temporal Alignment.** Rain rate $R(t)$ is derived as a 5 km spatial average over reference gauges/radar pixels along each CML path. Timestamps are aligned to a common 15-minute grid; temporal offsets between $\tilde{A}_i(t)$ and $R_j(t)$ are corrected prior to regression.

2. **Unit Alignment.** All rain rates are expressed in $\mathrm{mm\,h^{-1}}$ and attenuation in dB. Link length L is fixed in km per CML.

3. **CML Selection.** Analysis begins with a homogeneous subset of CMLs sharing the same frequency band. This controls for ITU coefficient variability across frequencies.

4. **Rain Event Definition.** Rain events are identified as periods containing either (i) a fraction $> \tau_f$ of rainy 15-minute samples, or (ii) a minimum number of consecutive samples with $R(t) > \tau_R \,\mathrm{mm\,h^{-1}}$. The threshold τ_R is applied to the reference rain rate $R(t)$, not to $\tilde{A}(t)$, since the latter requires baseline subtraction before it is physically interpretable.

5. **Data Split.** Data are partitioned at the link level into 80% training (PySR discovery and ITU coefficient fitting) and 20% held-out test (validation), preventing leakage from correlated per-link noise and hardware characteristics.

Step 2: Preprocessing — Baseline Estimation and Signal Isolation. Input: Raw observed attenuation $A_{\mathrm{obs}}(t)$ and aligned rain rate $R(t)$ for training links. **Output:** Cleaned attenuation signal $\tilde{A}(t)$ over rain-only samples, ready as PySR regression target.

1. **Baseline Estimation and Subtraction.** A dry-period baseline $A_{\mathrm{base}}(t)$ is estimated per event and subtracted to isolate weather-induced attenuation:
$$\tilde{A}(t) = A_{\mathrm{obs}}(t) - A_{\mathrm{base}}(t). \tag{23}$$

2. **Rain-Sample Masking.** Only samples with $R(t) > \tau_R$ are retained. Masking is applied on $R(t)$, not $\tilde{A}(t)$, to avoid circular thresholding before baseline correction is complete.

3. **Outlier Removal.** Physically implausible values (e.g. $\tilde{A}(t) < 0$ after baseline subtraction, or extreme spikes) are removed.

4. **WAA Correction (Stage 1b only).** The SRB wet-antenna estimate $\hat{A}_w^{\mathrm{SRB}}(t)$ is subtracted to yield the rain-only target:
$$A_r(t) = \tilde{A}(t) - \hat{A}_w^{\mathrm{SRB}}(t). \tag{24}$$
Stage 1a skips this step, using $\tilde{A}(t)$ directly.

Step 3: Experiments with Symbolic Regression via PySR. Input: Training pairs $\{R(t),\ \tilde{A}(t)\}$ (Stage 1a) or $\{R(t),\ A_r(t)\}$ (Stage 1b), with link length L per CML. **Output:** Pareto-optimal expressions ranked by complexity and MSE, compared against the ITU benchmark aLR^b.

E1 Specific Attenuation Discovery.
Target: $\tilde{A}(t)/L$. *Features:* $R(t)$.
PySR discovers the specific attenuation $g(R)$ such that:
$$\frac{\tilde{A}(t)}{L} \approx g(R(t)), \tag{25}$$

assuming ITU's linearity in L holds. Normalising by L isolates a single physical quantity (dB/km), giving the cleanest regression target directly comparable to the ITU-R specific attenuation $k = aR^b$.
Goal: Recover the functional form of specific attenuation and test whether $g(R) = aR^b$ is the correct structure (RQ1).

E2 Joint Discovery of $f(R, L)$.
Target: $\tilde{A}(t)$. *Features:* $R(t),\ L$.
PySR discovers the joint expression:

$$\tilde{A}(t) \approx f(R(t),\ L), \tag{26}$$

without constraining the role of L. Recovery of $f(R, L) = g(R) \cdot L$ confirms the ITU linearity assumption, while deviations such as L^γ indicate path-length effects beyond the ITU model.
Goal: Most general experiment; test whether PySR jointly recovers the ITU structure aLR^b or a more complex dependence on R and L. Requires sufficient variance in L across the dataset to be identifiable.

E3 Path-Length Dependence Isolation.
Target: $\tilde{A}(t)/\hat{a}R(t)^{\hat{b}}$. *Features:* L.
The ITU rain contribution is residualised using coefficients $\hat{a}, \hat{b}$ fitted from E1, leaving:

$$\frac{\tilde{A}(t)}{\hat{a}R(t)^{\hat{b}}} \approx f(L), \tag{27}$$

so that PySR discovers the L-dependence in isolation. Recovery of $f(L) = L$ confirms ITU linearity; deviations such as $f(L) = L^\gamma$ or a saturating form reveal non-linear path-length effects.
Goal: Targeted test of the ITU assumption $A \propto L$, decoupled from the rain functional form.

5.4.3 (Todo) Goal 2: Temporal Attenuation Dynamics via SINDy.

Goal 1 identifies the static relationship $\kappa(t) \approx \hat{a}R(t)^{\hat{b}}$, where $\kappa(t) = \tilde{A}(t)/L$ is the specific attenuation [dB/km]. However, the observed $\kappa(t)$ may exhibit temporal memory beyond this static mapping, arising from physical processes such as antenna wetting and drying, storm advection, and nonlinear rain-antenna interactions. We define the dynamic residual:

$$\varepsilon(t) = \kappa(t) - \hat{a}R(t)^{\hat{b}}, \tag{28}$$

which isolates temporal structure not explained by the static rain relationship. SINDy is applied per link to discover the governing ODE:

$$\dot{\varepsilon}(t) = \boldsymbol{\xi}^\top \Theta(\varepsilon,\ R,\ \dot{R}), \tag{29}$$

Table 2: PySR experiments for Goal 1. All experiments are repeated for Stage 1a ($\tilde{A}$) and Stage 1b (A_r), yielding a 3×2 design.

Exp.	Target	Features	Discovered form	Goal
E1	$\tilde{A}(t)/L$	$R(t)$	$g(R)$	Discover specific attenuation law; test whether $g(R) = aR^b$ is the correct functional form (RQ1).
E2	$\tilde{A}(t)$	$R(t),\ L$	$f(R, L)$	Most general; test whether PySR recovers aLR^b or a more complex R- and L-dependence.
E3	$\tilde{A}(t)/\hat{a}R^{\hat{b}}$	L	$f(L)$	Isolate path-length dependence; test whether $f(L) = L$ or a non-linear form (e.g. L^γ).
1a/1b	Repeat with $A_r(t) = \tilde{A}(t) - \hat{A}_w^{\mathrm{SRB}}(t)$			Quantify WAA bias; test whether correction simplifies discovered expressions (RQ2).

Research Questions.

RQ3 Is the static relationship of Goal 1 sufficient to explain $\kappa(t)$, or does SINDy identify significant residual dynamics $\dot{\varepsilon}(t)$?

RQ4 Which terms does SINDy select from Θ, and are they consistent with known physical mechanisms such as WAA memory or storm advection lag?

RQ5 Do the discovered per-link dynamics $\boldsymbol{\xi}$ vary systematically with link metadata (L, f), suggesting a path toward a semi-universal dynamic model?

5.5 Implementation.

Step 1: Specific Attenuation. **Input:** $\tilde{A}(t)$, L. **Output:** $\kappa(t) = \tilde{A}(t)/L$ [dB/km].

Step 2: Dynamic Residual. **Input:** $\kappa(t)$, $R(t)$, coefficients $\hat{a}, \hat{b}$ from Goal 1. **Output:** $\varepsilon(t) = \kappa(t) - \hat{a}R(t)^{\hat{b}}$.

Step 3: Numerical Differentiation. **Input:** $\varepsilon(t)$, $R(t)$. **Output:** $\dot{\varepsilon}(t)$, $\dot{R}(t)$.
A smoothing filter is applied prior to differentiation to suppress noise at 15-minute resolution.

Step 4: Candidate Library. **Input:** $\varepsilon(t)$, $R(t)$, $\dot{R}(t)$. **Output:** $\boldsymbol{\Theta} \in \mathbb{R}^{T \times p}$:

$$\Theta = \left[1,\ \varepsilon,\ \varepsilon^2,\ R,\ R^2,\ \dot{R},\ \varepsilon \cdot R,\ \sqrt{R}\right]. \tag{30}$$

Step 5: SINDy. **Input:** $\dot{\varepsilon}(t)$, $\boldsymbol{\Theta}$. **Output:** Sparse $\boldsymbol{\xi}$ per link.

Step 6: Post-Discovery Analysis. **Input:** Per-link $\boldsymbol{\xi}$, metadata L, f. **Output:** Consistency of discovered ODEs across links; correlation of $\boldsymbol{\xi}$ with L and f.

Link Metadata. SINDy is run independently per link. Links are stratified by frequency band f prior to analysis. Post-discovery, variability in $\boldsymbol{\xi}$ across links is examined for dependence on L and f, providing insight into whether discovered dynamics are universal or link-specific.

PySR–SINDy Integration. The two frameworks are complementary by construction. PySR (Goal 1) discovers the static relationship $\hat{f}(R)$, and SINDy (Goal 2) operates on the dynamic residual unexplained by it:

$$\kappa(t) = \underbrace{\hat{f}(R(t))}_{\text{PySR: static}} + \underbrace{\varepsilon(t)}_{\text{SINDy: dynamic}}, \tag{31}$$

where $\hat{f}$ is the expression selected from the PySR Pareto front, and $\varepsilon(t)$ captures temporal dynamics beyond the static rain relationship. This decomposition ensures the two goals are non-redundant and jointly provide a complete data-driven description of $\kappa(t)$.

6 Appendix *I*: Continuous-Time Modeling of Wet-Antenna Attenuation

Wet-Antenna Attenuation (WAA) represents a significant source of bias in rainfall retrieval from microwave links. To model the dynamic behavior of water accumulation and evaporation on the antenna radome, we synthesize empirical observations and discrete algorithms from the literature into a unified continuous-time framework.

6.1 Universal WAA model

Developing a universal deterministic model for Wet-Antenna Attenuation (WAA) is practically infeasible because attenuation is highly sensitive to chaotic, link-specific physical variables such as radome material coatings, surface degradation, and the antenna's specific elevation angle [**?**, **?**]. Furthermore, unpredictable environmental factors like differential wetting between the transmitting and receiving antennas, as well as whether water accumulates as individual droplets

or a continuous sheet, force researchers to rely on localized empirical calibration rather than explicit physical equations [**?**, 6].

6.2 Existing Models and Theoretical Background

Existing dynamic WAA models primarily treat the wetting and drying phases as distinct physical processes. Schleiss et al. (2013) proposed a discrete-time recursive model where the WAA during a rain event increases toward a maximum bound $A_{\max}$ (or W). Their algorithm updates the WAA at discrete intervals Δt_i according to:

$$w_i = w_{i-1} + (A_{\max} - w_{i-1})\frac{3\Delta t_i}{\tau_w}, \tag{32}$$

where τ_w is the time required to reach 95% of the maximum attenuation.

Conversely, for the drying phase, Minda and Nakamura (2005) empirically established that WAA decays exponentially as a continuous function of elapsed time once rainfall ceases, independent of path-attenuation measurements.

6.3 Derivation of the Continuous-Time Formulation

To facilitate continuous-time analysis, we adapt the discrete wetting step in (32) into a continuous Ordinary Differential Equation (ODE). By rearranging the terms and taking the limit as the time step approaches zero ($\Delta t \to 0$), the discrete update is transformed into the following first-order linear ODE:

$$\frac{dA_w(t)}{dt} = \frac{1}{\tau_{\text{wet}}}\left(A_{\max} - A_w(t)\right), \tag{33}$$

where τ_{wet} represents the standard mathematical time constant (the time to reach $1 - e^{-1}$, or $\approx 63.2\%$ of $A_{\max}$). Solving this differential equation for a rain event starting at t_0 with an initially dry antenna ($A_w(t_0) = 0$) yields the exact continuous-time exponential wetting formula:

$$A_w(t) = A_{\max}\left(1 - e^{-(t-t_0)/\tau_{\text{wet}}}\right). \tag{34}$$

For the drying phase, we directly adopt the continuous-time physical decay model proposed by Minda and Nakamura. During the drying phase starting at t_1, the accumulated water evaporates, yielding the exponential decay:

$$A_w(t) = A_w(t_1)\, e^{-(t-t_1)/\tau_{\text{dry}}}, \tag{35}$$

where τ_{dry} is the drying time constant.

6.4 Model Assumptions and Approximations

By adopting the formulations in (34) and (35), this model locks in several specific physical approximations regarding the antenna's thermodynamics and hydrodynamics:

- **Time-Driven Wetting Limit:** The model assumes that $A_w(t)$ wets at a predictable exponential rate strictly as a function of elapsed time $(t - t_0)$. This approximates the wetting process by isolating it from the instantaneous variations in rainfall intensity, unlike models that tie WAA directly to the specific rain rate or total path attenuation.
- **Constant Environmental Drying:** The parameter τ_{dry} is treated as a constant. This assumes a uniform exponential evaporation rate, effectively averaging out the real-world meteorological variables (e.g., wind speed, ambient temperature, and relative humidity) that dictate the actual thermodynamic drying process.
- **Static Saturation Bound:** The maximum attenuation $A_{\max}$ is assumed to be a fixed upper bound for a given antenna and frequency. This assumes that once the radome surface is completely covered by a water film or droplet distribution, further rainfall does not significantly increase the attenuation.

Table 3: Summary of SINDy/PySR discovery goals for CML attenuation analysis.

Goal	Objective	Method	Validation
1	Baseline removal	Preprocessing	Visual / standard CML
2	Power-law discovery	PySR	ITU coefficients
3	WAA dynamics discovery	SINDy	vs. SRB model
4	Joint dynamic model	SINDy-c	Forward simulation
5	Rainfall estimation	Model inversion	vs. PWS ground truth

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